

Introduction to **CSS**

HTML course: Lesson 5



Lesson Plan

1

What is CSS and what does it consist of

2

What are selectors, properties, and their values

3

How to link styles to a page

4

Viewport and units, colors

5

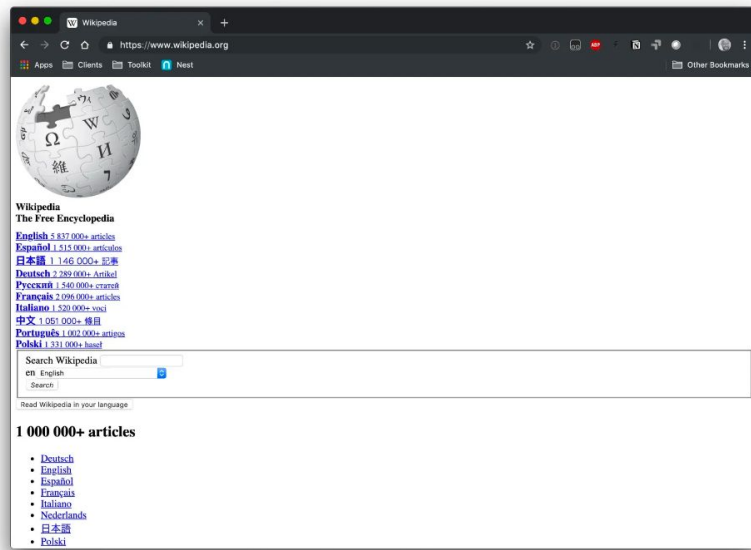
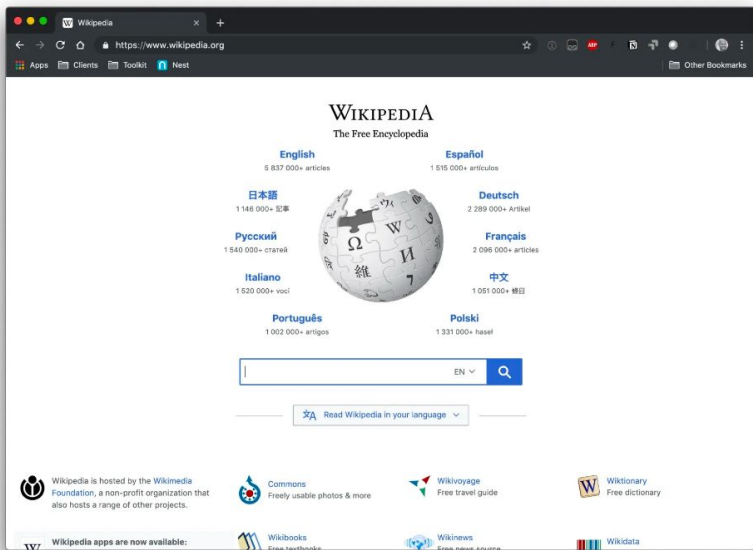
How to work with fonts

Cascading Style Sheet

CSS level 3



Describes how elements should be displayed on various media such as screens, paper, speech, or other media.



CSS

Consists of Rules:

Selectors are expressions that tell the browser which HTML element to apply the CSS properties defined within the style block.

Inside a **rule**, **selectors** and **declaration blocks** are defined, consisting of **properties** and their **values**.



CSS

shorthand properties

```
.box {  
  padding-top: 5px;  
  padding-right: 10px;  
  padding-bottom: 10px;  
  padding-left: 5px;  
}
```

That's **92 characters**, totaling **92 bytes**.

```
.box {  
  padding: 5px 10px 10px 5px;  
}
```

That's **37 characters**, totaling **37 bytes**.

- background
- font
- border
- outline
- margin
- padding
- list

Types of

Selectors

- Tag Selector

`h1 {}`

- Class Selector

`.main-heading {}`

- ID Selector

`#reasons-section {}`

- Attribute Selector

`a[href] {}`

- Pseudo-classes

`a:hover {}`

- Pseudo-elements

`p::first-line {}`

- Universal Selector

`* {}`

- Grouped Selectors

`h1, .heading {}`

Selector Combinators

descendant combinator	space	<code>body article p</code>	elements are nested inside each other	<pre><body> <article> <p></p> </article> </body></pre>	<pre><body> <section> <p></p> </section> <p></p> </body></pre>
child combinator	<code>></code>	<code>article > p</code>	an element is an immediate child	<pre><article> <p></p> <a> </article></pre>	<pre><article> <div> <p></p> </div> <p></p> </article></pre>
adjacent sibling combinator	<code>+</code>	<code>p + img</code>	an element that comes immediately after	<pre><article> <p></p> <a> </article></pre>	<pre><article> <p></p> <a> </article></pre>
general sibling combinator	<code>~</code>	<code>p ~ img</code>	an element located anywhere below in the code, inside a common parent	<pre><article> <p></p> <a> </article></pre>	<pre><article> <p></p> <a> </article></pre>

Selector

Combinations

<code>p { }</code>	<pre><p class="intro module"></p> <div class="intro"></div> <p class="default-text"></p></pre>
<code>p a { }</code>	<pre><p> <div> <a> <a> </p> </div></pre>
<code>.intro { }</code>	<pre><p class="intro"></p> <div class="intro"></div></pre>
<code>div.intro { }</code>	<pre><p class="intro"></p> <div class="intro"></div></pre>
<code>.intro.module { }</code>	<pre><p class="module"></p> <p class="module intro"></p> <p class="intro module"></p> <p class="intro"></p></pre>

<code>.intro .module { }</code>	<pre><p class="intro module"></p> <div class="intro module"> <p class="module"></p> </div></pre>
<code>.intro, .module { }</code>	<pre><p class="intro"></p> <div class="module"></div></pre>
<code>.intro { } ≠ .Intro { }</code>	<pre><p class="intro"></p> <div class="Intro"></div></pre>
<code>.nav ul li a { }</code>	<code>.nav a { }</code>
<code>/* comment */</code>	
Selectors are read from right to left <code>nav ul li { }</code>	The rule applies to the <code>li</code> element, which is part of the <code>ul</code> element, which is inside the <code>nav</code> element.

Applying Styles in Order

1

Inheritance

2

Cascading

3

Specificity

1

Browser Styles

2

User Styles

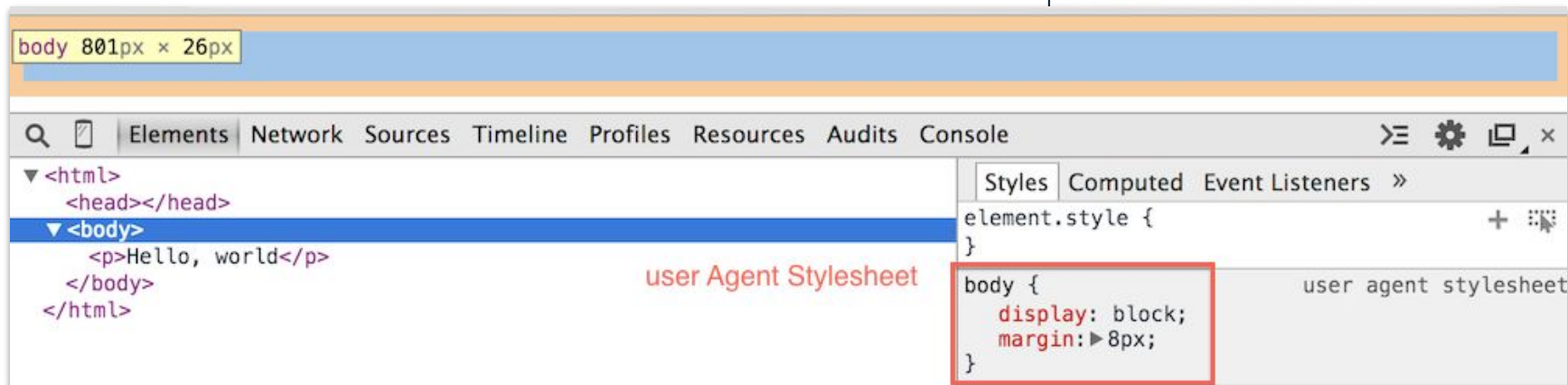
3

Author Styles

Default Browser Styles

User agent stylesheet

– styles present in any browser.



Способ приготовления

Шаг 1



Как получить миндальную муку:

- Миндальная мука можно приготовить следующим образом:
- В магазине, где продают орехи можно попросить, чтобы их смолотли в муку (они это делают в кофемолке).
- Просеять ее.
- Миндаль должен быть очищенным, если вы делаете белые макарруны.
- Если шоколадные макарруны, то можно миндаль брать обычный.

Как делать муку для макаррун:

1. Нагреть духовку до 150 градусов.
2. Смешать миндальную муку, сахарную пудру и какао в комбайне, 2 минуты.
3. Противень застелить бумагой для выпечки высыпать сухую смесь на лист, просушить в духовке в течение 5 минут.
4. Просеять через очень мелкое сито.

Шаг 2

- A. Приготовить ганаш.
- B. Сливки подогреть, добавить поломанный на кусочки шоколад.
- C. Размешать до полного растворения.
- D. Дать остыть и поставить в холодильник (желательно на ночь).

User styles

The user can customize fonts, colors, positions of links in the margins, and many other things.

The image shows a YouTube interface with a custom theme. The background is pink with white and yellow floral patterns. The video player shows a Hello Kitty character with a large red bow. A blue menu is overlaid on the video, listing options: Loop, Copy video URL, Copy video URL at current time, Copy embed code, Copy debug info, Troubleshoot playback issue, and Stats for nerds. The video title is "Hello Kitty and Friends Supercute Adventures: The Perfect Gift (NEW ANIMATION) S1 EP 1". The channel name is "Hello Kitty and Friends" with 607K subscribers. The video has 56K likes. On the right, a user menu is open for the user "se7en", showing options like "Manage your Google Account", "Your channel", "YouTube Studio", "Switch account", "Sign out", "Purchases and memberships", "Your data in YouTube", "Appearance: Light", "Language: English", "Restricted Mode: Off", "Location: Philippines", "Keyboard shortcuts", "Settings", "Help", and "Send feedback".

YouTube PH

Search

Loop

Copy video URL

Copy video URL at current time

Copy embed code

Copy debug info

Troubleshoot playback issue

Stats for nerds

1:37 / 3:42

#HelloKittyandFriends #SupercuteAdventures #Animation

Hello Kitty and Friends Supercute Adventures: The Perfect Gift (NEW ANIMATION) S1 EP 1

Hello Kitty and Friends
607K subscribers

Subscribe

56K

Share

Save

se7en

Manage your Google Account

Your channel

YouTube Studio

Switch account

Sign out

Purchases and memberships

Your data in YouTube

Appearance: Light

Language: English

Restricted Mode: Off

Location: Philippines

Keyboard shortcuts

Settings

Help

Send feedback

Author Styles

Site styles.

Top values for you

Try variety of benefits when using our services



Airport pickup

We provide escort from the airport to the hotel



Easy booking

Quick and easy booking of tours for upcoming dates



Best tour guide

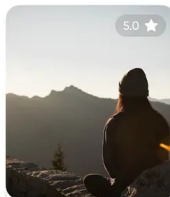
Our best tour guide is ready to guide your trip



Lots of promos

Various promotions and drawings of tours

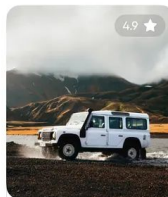
Choose your tour



5.0 ★

Alone with nature

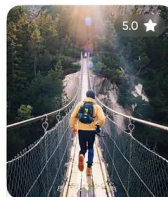
\$100/person



4.9 ★

Jeep ride

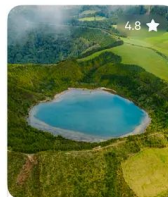
\$200/person



5.0 ★

Hiking tour

\$150/person



4.8 ★

Corners of the island

\$220/person



See all

Enjoy your travel

Discover the most unique and vibrant experiences the Azores have to offer

Lighthouse Farol
Ponta do Arnel

Enjoy your travel

Discover the most unique and vibrant experiences the Azores have to offer

Lighthouse Farol
Ponta do Arnel

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Weather Nature Azores



Beautiful nature

Beaches



Hiking in the mountains

Trails Nature Beaches

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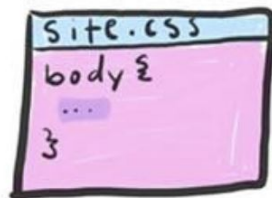
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Health & safety
Privacy policy

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Facebook Instagram Twitter



user
agent



author
styles



local
user



base

base

base

!important

!important

!important

least
specific



most
specific

Reset and Normalize CSS

normalize.css

reset.css

- Maintains useful browser settings
- standardizes styles
- corrects errors and inconsistencies
- improves usability
- includes comments and documentation.

Connects to your style file:

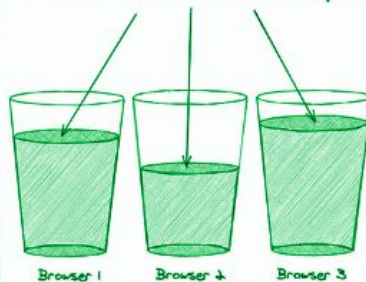
```
<link rel="stylesheet" type="text/css"
href="https://necolas.github.io/normalize.css">
```

Visualizing CSS Resets

Resets bring sanity when dealing with cross-browser style inconsistencies.

Imagine that browsers were glasses and default styles were water...

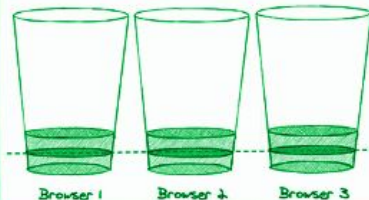
Each browser applies a slightly different set of default styles



CSS Reset

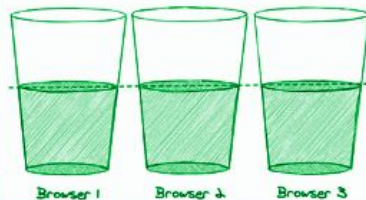
Removes most styles requiring devs to add consistent styles

REMAINING STYLES DEV ADDED STYLES



CSS Normalize

Removes only inconsistent styles and keeps as many common styles as possible



elijahmanor.com/css-resets

@elijahmanor

Practice:

● What color will be applied to the header

```
/* user-agent style sheet */
```

```
h2 { color: black; }
```

```
/* user style sheet */
```

```
h2 { color: green; }
```

```
/* author style sheet */
```

```
h2 { color: blue; }
```

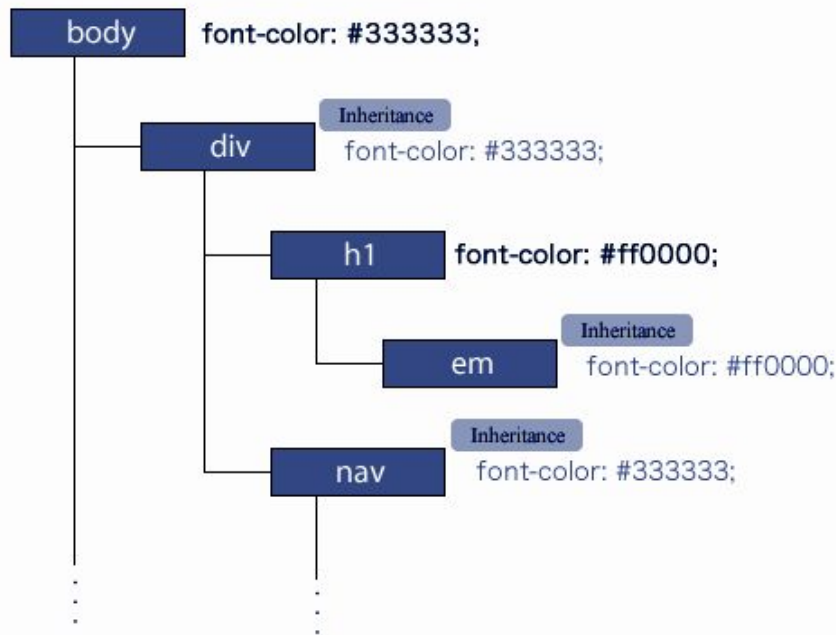
Applying Styles in Order

1	1
Inheritance	Browser Styles
2	2
Specificity	User Styles
3	3
Cascading	Author Styles

Style

Inheritance

- + Styles don't need to be applied to every element.
- + Styles don't need to define every property for each element.
- + CSS files can be smaller, easier to read, and load faster.
- + Only properties that simplify the developer's work are inherited.



Style

Inherited

color: value;

List related properties

list-style-image: value;
list-style-position: value;
list-style-type: value;
list-style: value;



– styles that need to be set for body to be inherited by all page elements

Text related properties

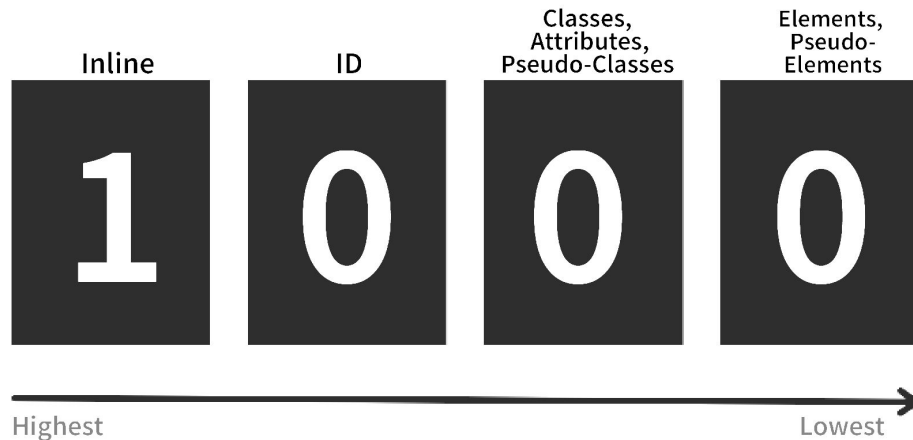
font-family: value;
font-size: value;
font-style: value;
font-variant: value;
font-weight: value;
font: value;
letter-spacing: value;
line-height: value;
orphans: value;
text-align: value;
text-indent: value;
text-transform: value;
widows: value;
word-spacing: value;

Specificity of selectors

Determines which styles will be applied to an element based on the selector's weight.

Avoid overusing weight; try to create lightweight selectors.

Selector nesting should be no more than three levels.



specificity = weight

universal selector * has no specificity weight (0)

	important	inline styles	id	class, attribute, pseudo-class	elements, pseudo-elements	
	10000	1000	100	10	1	
<code>.card a:hover</code>	0	0	0	10+10	1	21
<code>.card .heading::before</code>	0	0	0	10*2	1	21
<code>body #block > .card</code>	0	0	100	10	1	111
<code>.card .heading .highlight + a[href]</code>	0	0	0	10*3+10	1	error
<code>body main #block .card h2 ~ a</code>	0	0	100	10	1*4	114
<code>.default-link {...!important}</code>	10000	0	0	10	0	10010
<code></code>	0	1000	0	0	0	1000
<code></code>	10000	1000	0	0	0	11000

Specificity of selectors

Examples

```
<body>
  <main>
    <section id="block">
      <article class="card">
        <h2 class="heading">
          <span class="highlight"></span>
        </h2>
        <a class="default-link">
```

Practice: Calculating specificity

#header h1 span a { }

a =  x inline styles
b =  x ID
c =  x classes
d =  x elements
Specificity = 

.intro ::first-letter { }

a =  x inline styles
b =  x IDs
c =  x class
d =  x pseudo-element
Specificity = 

a[href^="http:"] { }

a =  x inline styles
b =  x IDs
c =  x attribute selector
d =  x element
Specificity = 

Practice: Calculating specificity

`#header h1 span a { }`

a = 0 x inline styles
b = 1 x ID (*#header*)
c = 0 x classes
d = 3 x elements (*h1,span,a*)
Specificity = 0,1,0,3

`.intro ::first-letter { }`

a = 0 x inline styles
b = 0 x IDs
c = 1 x class (*.intro*)
d = 1 x pseudo-element (*::first-letter*)
Specificity = 0,0,1,1

`a[href^="http:"] { }`

a = 0 x inline styles
b = 0 x IDs
c = 1 x attribute selector (*[href^="http:"]*)
d = 1 x element (*a*)
Specificity = 0,0,1,1

Practice:

● What color will be applied to the text

```
<div id="container">  
  <nav class="nav">  
    <p class="intro">
```

```
.nav p { color: yellow; }
```

```
p { color: blue; }
```

```
div#container p { color: purple; }
```

```
p.intro { color: orange; }
```

Practice:

● What color will be applied to the text

<code>.nav p { color: lime; }</code>	<code>0,0,1,1</code>
<code>p { color: blue; }</code>	<code>0,0,0,1</code>
<code>div#container p { color: purple; }</code>	<code>0,1,0,2</code>
<code>p.intro { color: green; }</code>	<code>0,0,1,1</code>

Cascading

Styles

When selectors have the same weight, the one defined later in the file takes precedence.

Hence, the name Cascading Style Sheets.

that **Cascading** Style Sheet



```
/*The final background color of the h1 element will be green*/  
h1 {  
  background-color: red;  
  font-size: 10px;  
}  
  
h1 {  
  background-color: green;  
}
```

Connecting Styles

Inline CSS

```
<p style="color: blue;">This is a paragraph.</p>
```

Internal CSS

```
<head>
  <style type = text/css>
    body {background-color: blue;}
    p { color: yellow;}
  </style>
</head>
```

External CSS

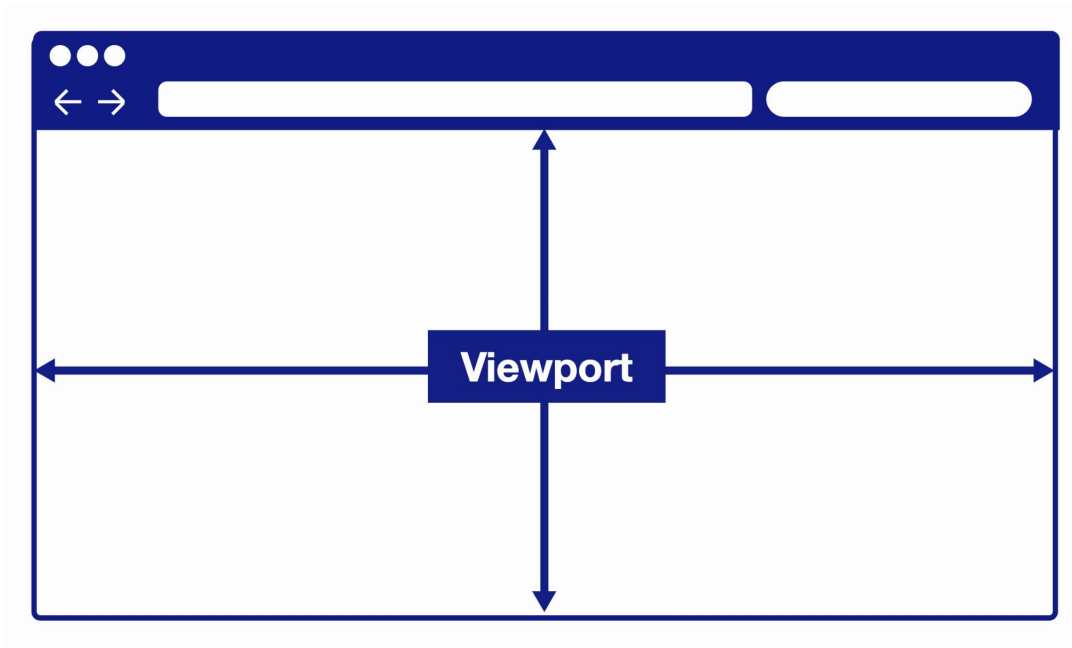
```
<head>
  <link rel="stylesheet" type="text/css" href="style.css">
</head>
```

- inline styles
- embedded (tag <style> in head)
- linked (tag <link>)
One stylesheet for all pages.

viewport

– is the visible area of the browser window, which varies depending on the device:

- Smartphone
- Tablet
- Computer



viewport units:

vh / vw

(% of viewport height/width)

Absolute

Units

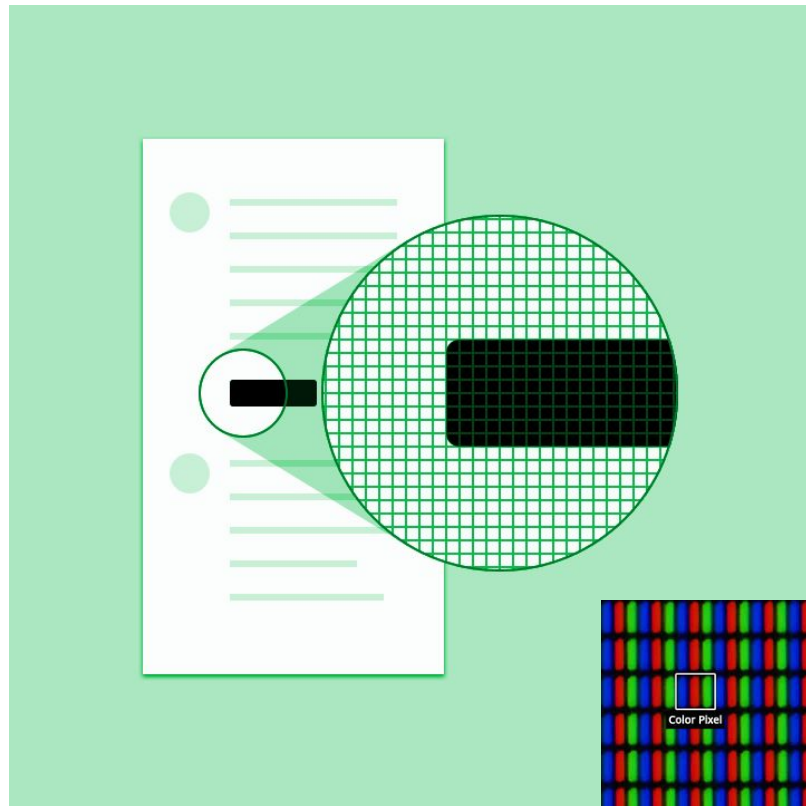
– fixed physical unit (1/96th of an inch (0.25mm))

px

`width: 100px;`
`height: 100px;`

`margin: 20px;`
`padding: 20px;`

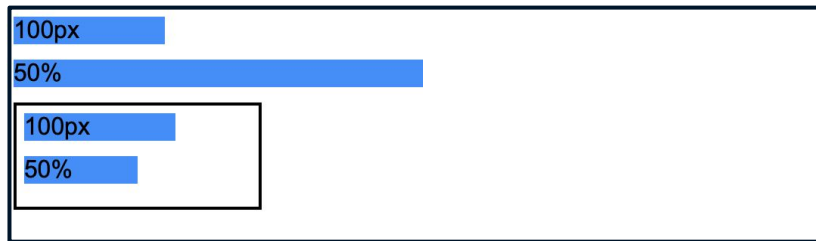
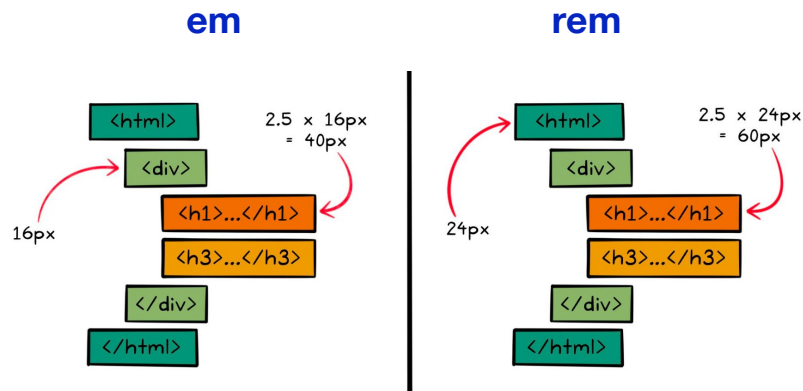
`font-size: 15px;`



Relative

Units

- **em** (from parent) / **rem** (from html tag)
 - **font-size: 1.5rem;**
 - **font-size: 1.2em;**
 - **margin: 0.5rem;**
 - **padding: 0.8em;**
- **%** (relative to parent element's property)
 - width: 50%;**
 - height: 75%;**
 - margin: 10%;**
 - padding: 5%;**



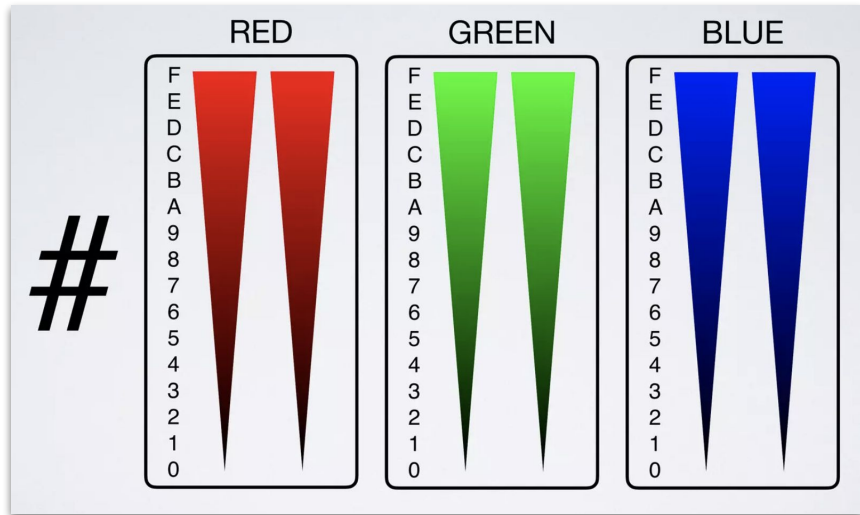


Colors in CSS

- hex / hexa
- rgb / rgba
- color keywords
- currentcolor
- transparent

Properties:

- color
- background-color
- text-shadow
- text-decoration-color
- border-color
- box-shadow
- outline



Color

HEX/HEXA

```
p {color: #FFAA00;}
```

```
p {color: #FA0;}
```

```
p {color: #FFAA0000;} <- full transparency
```

```
p {color: #FFAA00FF;} <- full opacity
```

consists of:

- digits from **0** to **9**
- letters from **A** to **F**
 - **0** – black color
 - **F** – full color
- hexadecimal system

Color

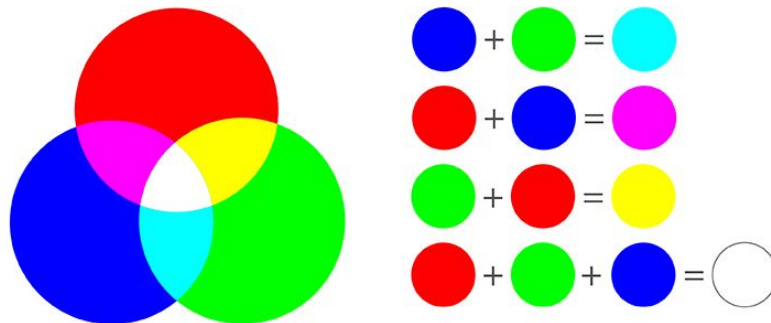
RGB/RGBA

```
p {color: rgb(0, 0, 0);}
p {color: rgb(100%, 100%, 100%);}
p {color: rgba(50%, 10%, 60%, 1);}
p {color: rgba(31, 250, 118, 0);}
```

consists of:

- digits from **0 (black)** to **255 (white)**
or
- from **0% (black)** to **100% (white)**
- transparency from **0 (transparent)**
to **1 (fully opacity)**

RGB



New Notation:

```
p { color: rgb(100% 100% 100%); }
p { color: rgb(255 255 255 / .5); }
p { color: rgb(0 0 0 / 50%); }
```

- no commas
- alpha channel is separated by a slash – /
- alpha channel can be in %
- not supported by older browsers

Color	Name	Hex	RGB
	aqua	#00FFFF	0, 255, 255
	black	#000000	0, 0, 0
	blue	#0000FF	0, 0, 255
	fuchsia	#FF00FF	255, 0, 255
	gray	#808080	128, 128, 128
	green	#008000	0, 128, 0
	lime	#00FF00	0, 255, 0
	maroon	#800000	128, 0, 0
	navy	#000080	0, 0, 128
	olive	#808000	128, 128, 0
	purple	#800080	128, 0, 128
	red	#FF0000	255, 0, 0
	silver	#C0C0C0	192, 192, 192
	teal	#008080	0, 128, 128
	white	#FFFFFF	255, 255, 255
	yellow	#FFFF00	255, 255, 0

Color

Keywords

white	gainsboro	silver	darkgray	gray	dimgray	black
whitesmoke	lightgray	lightcoral	rosybrown	indianred	red	maroon
snow	mistyrose	salmon	orangered	chocolate	brown	darkred
seashell	peachpuff	tomato	darkorange	peru	firebrick	olive
linen	bisque	darksalmon	orange	goldenrod	sienna	darkolivegreen
oldlace	antiquewhite	coral	gold	limegreen	saddlebrown	darkgreen
floralwhite	navajowhite	lightsalmon	darkkhaki	lime	darkgoldenrod	green
cornsilk	blanchedalmond	sandybrown	yellow	mediumseagreen	olivedrab	forestgreen
ivory	papayawhip	burlywood	yellowgreen	springgreen	seagreen	darkslategray
beige	moccasin	tan	chartreuse	mediumspringgreen	lightseagreen	teal
lightyellow	wheat	khaki	lawngreen	aqua	darkturquoise	darkcyan
lightgoldenrodyellow	lemonchiffon	greenyellow	darkseagreen	cyan	deepskyblue	mediumslateblue
honeydew	palegoldenrod	lightgreen	mediumaquamarine	cadetblue	steelblue	darkblue
mintcream	palegreen	skyblue	turquoise	dodgerblue	blue	darkblue
azure	aquamarine	lightskyblue	mediumturquoise	lightslategray	blueviolet	mediumslateblue
lightcyan	paleturquoise	lightsteelblue	cornflowerblue	slategray	darkorchid	darkslateblue
aliceblue	powderblue	thistle	mediumslateblue	royalblue	fuchsia	purple
ghostwhite	lightblue	plum	mediumpurple	slateblue	magenta	darkviolet
lavender	pink	violet	orchid	mediumorchid	mediumvioletred	purple
lavenderblush	lightpink	hotpink	palevioletred	deeppink	crimson	darkmagenta

Keyword

currentColor

– Inherits the **color** value from the element or parent.

```
div {  
  color: red;  
  border: 5px solid currentColor;  
}
```

The border color will be red.

```
div {  
  color: red;  
  border: 5px solid currentColor;  
  color: yellow;  
}
```

The border color will be yellow because of the cascade rule.

```
div {  
  color: green;  
}  
div p {  
  border: 5px solid currentColor;  
}
```

The border color will be green because the color is defined by the parent.



Useful for making the color of a vector icon match the text color.

Keyword

transparent

– Fully transparent.

```
div {  
  color: transparent;  
  border: 5px solid transparent;  
}
```

The text will take up space on the page but will not be visible. Its outline will also not be visible.

Other ways to set a fully transparent color:

```
color: rgba(0, 255, 0, 0);  
color: #FFFFFF00;
```

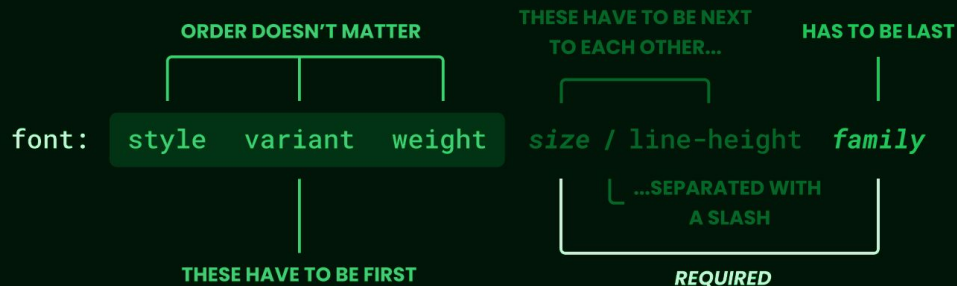
opacity: 0; – the entire element will be transparent

Working with

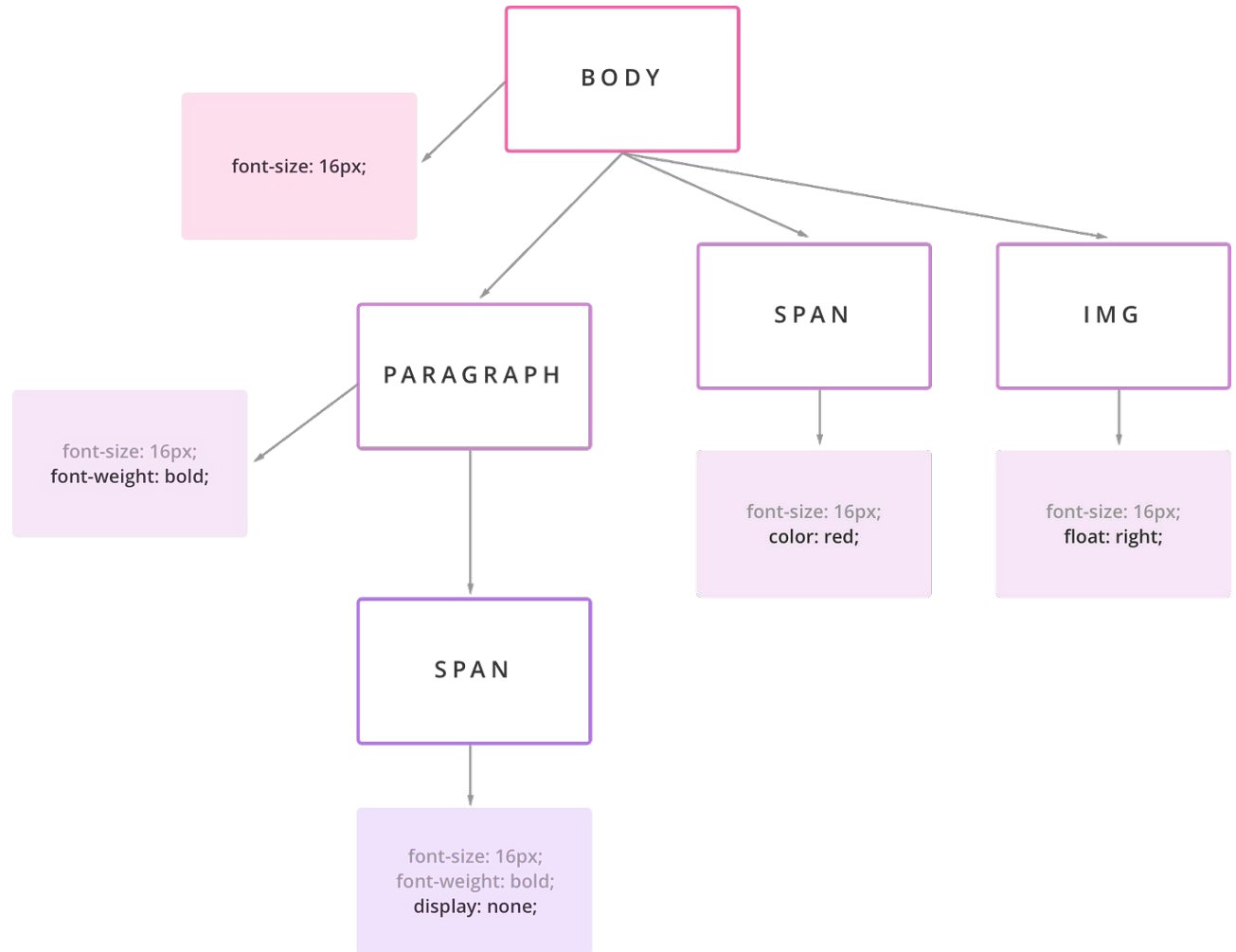
Text

- ***font-style***
- font-weight
- **font-size**
- **line-height**
- **font-family**
- **color**

```
body {  
  font-style: italic;  
  font-variant: small-caps;  
  font-weight: bold;  
  font-size: 16px;  
  line-height: 1.2em;  
  font-family: Arial, Helvetica, sans-serif;  
}
```



CSSOM



CSS

font-family

font family =

- A font from the design
- A web-safe font
- Font type

This ensures the text looks as similar to the layout as possible, even if the preferred font isn't loaded.

```
font-family: 'PT Sans',  
Arial, sans-serif;
```

Types of fonts:

- **Serif** – Fonts with small lines at the end of characters
- **Sans-serif** - Fonts without those lines
- *Cursive* – Handwriting-like fonts, often with connected letters
- **Monospace** - Fonts where all characters have the same width
- **Fantasy** – Decorative fonts

Web-safe

Fonts

Sans-serif

- Arial
- Verdana
- Tahoma
- Trebuchet MS

Serif

- Times New Roman
- Georgia
- Garamond

- Courier New (**monospace**)
- Brush Script MT (**cursive**)

```
font-family: "Inter", Arial, sans-serif;
```

```
font-family: "Playfair Display", Georgia, serif;
```

Connecting a Font

1. Open fonts.google.com
2. **Search** for the desired font
3. Choose the desired styles
4. In the side panel, select the **"link"** option
5. Copy the **code and paste** it into the `<head>` tag in the HTML
6. Set the font for the page in the **CSS file**

The screenshot shows the Google Fonts website interface. At the top, a search bar contains the text 'Inter'. Below the search bar, there are filters for 'Categories', 'Language', and 'Number of styles'. A notification indicates '21 of 1562 families'. The main display shows the 'Inter' font family by 'Rasmus Andersson', labeled as 'Variable (2 axes)'. The text 'Whereas recognition of the inherent dignity' is shown in the 'Inter' font. A side panel on the right shows the 'Use on the web' section, which includes a 'link' option selected. Below this, the HTML code for embedding the font is displayed:

```
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;500&display=swap" rel="stylesheet">
```

 Below the code, the CSS rule to specify the font family is shown:

```
font-family: 'Inter', sans-serif;
```

 The bottom part of the screenshot shows the font preview with different weights (Light 300, Regular 400, Medium 500) applied to the same text. The 'Regular 400' weight is selected.

2

Inter

Sentence Type something

Categories Language Number of styles Show only variable fonts Show only color fonts

21 of 1562 families

Inter
Rasmus Andersson

Variable (2 axes)

Whereas recognition of the inherent dignity

2.1

Akatar SIL Inter

Use on the web

To embed a font, copy the code into the `<head>` of your html

4

☒ `<link>` ☐ `@import`

```
<link rel="preconnect" href="https://
fonts.googleapis.com">
<link rel="preconnect" href="https://
fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.
com/css2?family=Inter:wght@300;500&di
splay=swap" rel="stylesheet">
```

5

CSS rules to specify families

```
font-family: 'Inter', sans-serif;
```

6

Light 300

Whereas recognition of the inherent dignity

Remove Light 300

Regular 400

Whereas recognition of the inherent dignity

Select Regular 400

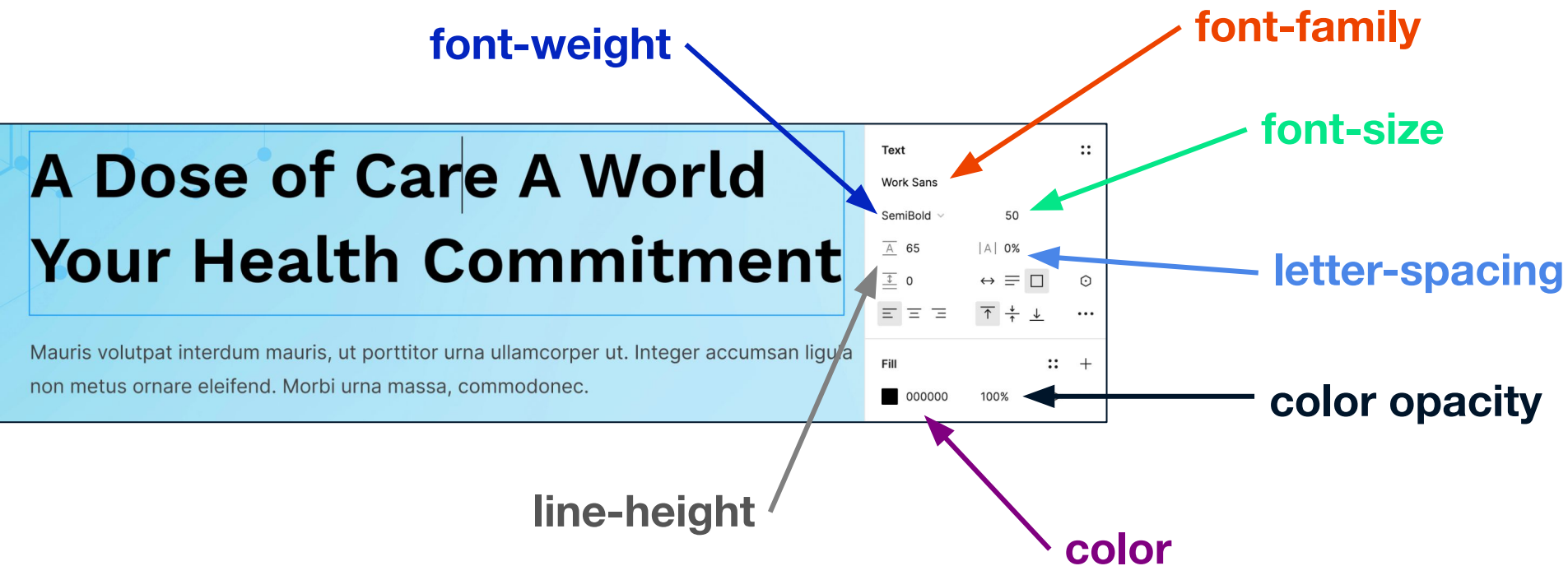
Medium 500

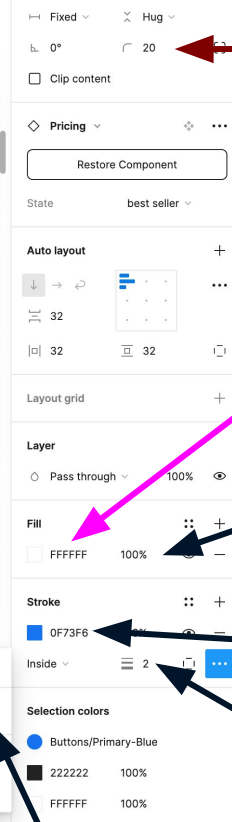
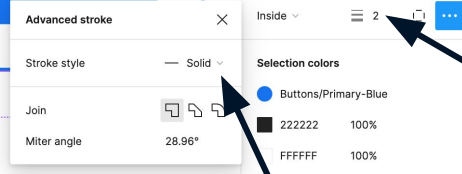
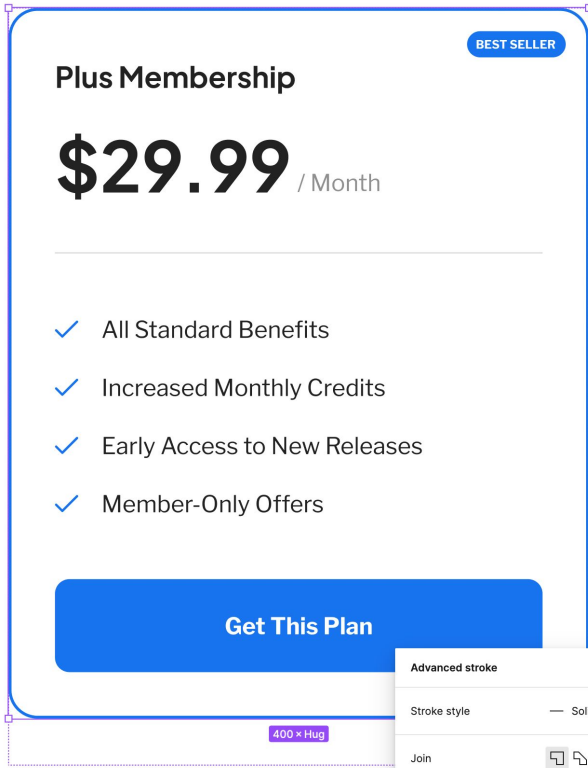
Whereas recognition of the inherent dignity

3

Remove Medium 500

Styles in Figma





border-radius

background-color

background-color opacity

border color

border width

border style

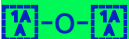
Vendor Prefixes

— Special prefixes added to CSS properties to allow developers to use experimental or new CSS properties before they become standard and are supported by all browsers.

WebKit 

(Safari, old versions of Chrome)

Mozilla Firefox 

Opera 

Internet Explorer и Microsoft Edge 

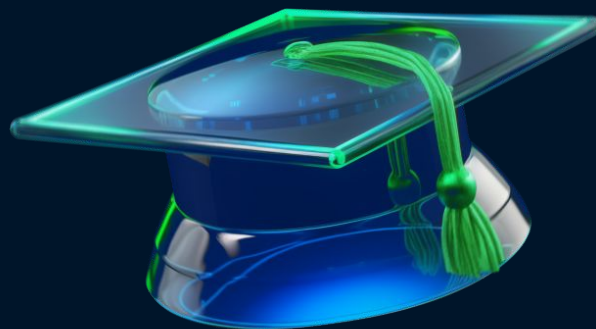
(old versions)

```
background-clip: text;  
-webkit-background-clip: text;  
-webkit-text-fill-color: transparent;
```

Use Autoprefixer [Autoprefixer](#).

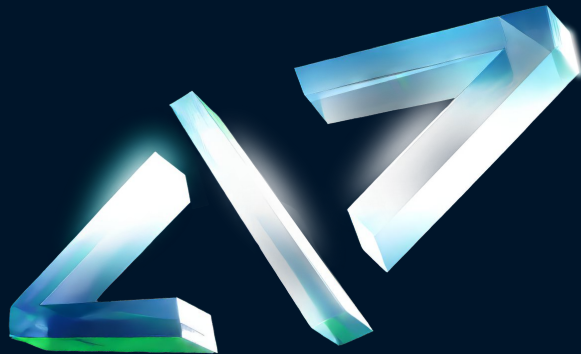
Summary

1. CSS syntax
2. What selectors are
3. The order in which styles are applied
4. How to link styles to a page
5. The purpose of a style normalizer
6. Units of measurement and colors used in CSS
7. How to link fonts



Homework

1. Complete one of the courses to reinforce the theory.
2. Connect ``normalize.css`` to Binabox project as a separate file.
3. Add a font using Google Fonts.
4. Write basic styles for the marked-up blocks of the layout using Emmet right away:
 - font size / Line height / Font type / Font weight - font-*
 - Text color
 - Background color



Quality Criteria for HTML Course

❤️ Mandatory for passing the course

💛 Required for the highest grade

💚 Optional

❤️ Single CSS File.

❤️ All fonts used in the design are connected to the pages.

❤️ Provide fallback fonts and family types at the end of the font list.

❤️ Do not use !important in CSS.

❤️ Do not use #id for styling.

❤️ Use consistent units for element sizes and positioning.

💛 Avoid nesting selectors more than two levels deep.

💛 Avoid styling tags directly

💛 Colors should be in a consistent format (hex or rgba).

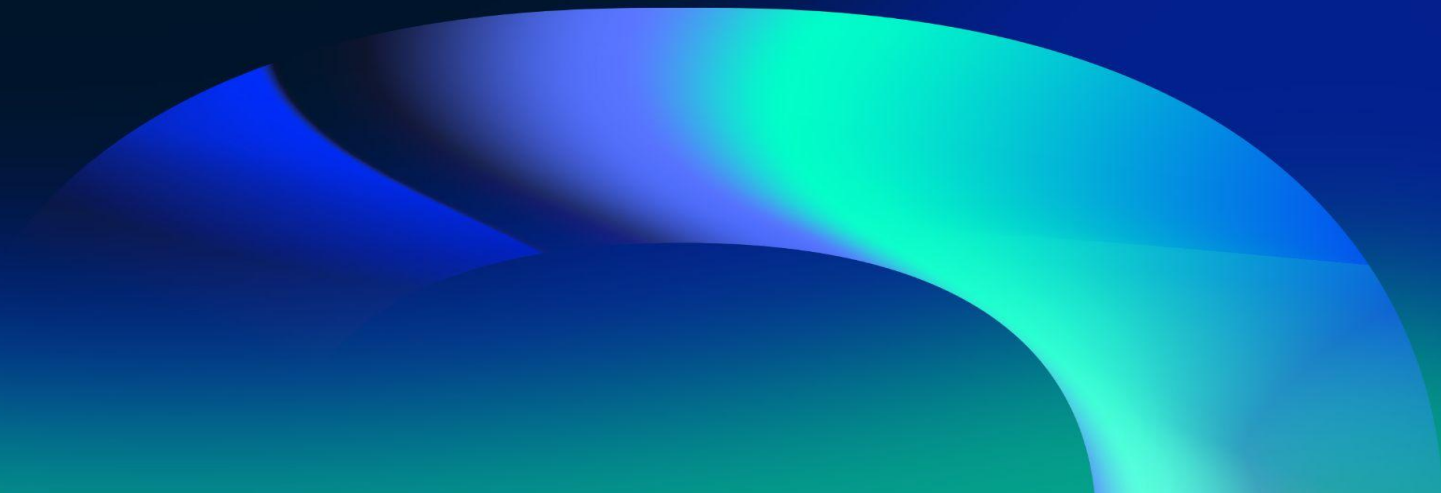
💚 Include Normalize.css

B Academy
RO



QUESTIONS?

Please fill out the feedback form
It's very important for us





THANK YOU!

Have a good evening!